

It measures excess return per unit of risk, helping you understand if your returns are due to smart investment decisions or just excessive risk-taking. Think of the Sharpe Ratio as your trading strategy's "efficiency score." A higher Sharpe Ratio indicates that your strategy is generating more return for each unit of risk taken—like a car that gets more miles per gallon.

The formula is:

$$\text{Sharpe Ratio} = (\text{Strategy Return} - \text{Risk-Free Return}) / \text{Strategy Standard Deviation}$$

In Python, we can calculate it like this:

```
1. # Assuming we have daily strategy returns in a Series called 'strategy_returns'
2. risk_free_rate = 0.02 / 252 # 2% annual risk-free rate converted to daily
3.
4. excess_returns = strategy_returns - risk_free_rate
5. sharpe_ratio = np.sqrt(252) * (excess_returns.mean() / excess_returns.std())
6.
```

A Sharpe Ratio above 1.0 is considered acceptable, above 2.0 is good, and above 3.0 is excellent. Many professional trading strategies target a Sharpe Ratio of at least 1.5.

?? **Fun Fact:** When Renaissance Technologies was recruiting quantitative analysts in the 1990s, they would ask candidates to estimate the Sharpe ratio of various gambling games like blackjack and roulette. Their founder, Jim Simons, believed that understanding the risk-adjusted return of these games helped identify minds that could develop profitable trading strategies. This approach helped them build one of the most successful hedge funds in history, with their Medallion Fund achieving a Sharpe ratio reportedly above 4.0—an almost unheard of level of risk-adjusted performance. For comparison, most professional hedge funds consider a Sharpe ratio of 1.0 to be acceptable and 2.0 to be excellent.